

Real-time rendering of underwater scenes based on data and an approximation to multiple scattering

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Abstract

We propose a physically-based, multispectral simulation to render underwater scenarios in real time, which also takes into account the RGB response curve of arbitrary sensors. Underwater illumination is mostly governed by multiple scattering, where light is scattered a number of times between particles before reaching the sensor. This phenomenon is therefore very low frequency and can be modeled as a function of depth and wavelength. Our approximation to multiple scattering is based on the measurable coefficient of diffuse downwelling attenuation. We combine this approximation with the standard froxel lighting technique for single scattering, and show examples simulating underwater appearance under different scattering, absorption and downwelling coefficients of the Jerlov water types.

1. Introduction

Spectral rendering, where more than the three RGB wavelength bands are considered, is especially well suited for underwater scenarios. The more common atmospheric image formation model is not appropriate to simulate the appearance of the ocean, since the absorption coefficient (often more prevalent than scattering) shows great wavelength dependency [AT18].

We present a spectral approximation to low frequency underwater illumination suitable for real-time rendering, capable of reproducing a large variety of physically-based scenarios, such as the different Jerlov water types [SM15], or different depths. We also combine this illumination with the standard froxel lighting technique for high frequency single scattering.

Participating media Illumination in participating media such as water is more complex than surfaces, as every point along the ray contributes towards the illumination reaching the eye [GJJD09]. On non-emissive media, there are only two coefficients that model this behaviour. The scattering coefficient, σ_s (m^{-1}) is the differential probability of light being scattered per unit length travelled along the medium, while the absorption coefficient σ_a (m^{-1}) represents the probability of absorption. These are combined into extinction $\sigma_t = \sigma_a + \sigma_s$, which fully accounts for radiance reduction events.

Radiative Transfer Equation The radiative transfer equation (RTE), introduced by Chandrasekhar in 1950 [Cha50], describes this interaction between light and media in a differential manner. In computer graphics, it is common to formulate the RTE in its integral form, also known as the volume rendering equation (VRE) [FWKH17]. This expression accounts for the final radiance

$L(\mathbf{x}_S, \omega)$ reaching the camera from $\mathbf{x}_S$, traveling S units in direction ω , so it will allow us to render any homogeneous underwater medium given its physical parameters:

$$L(\mathbf{x}_S, \omega) = T(\mathbf{x}_S) L_s(\mathbf{x}_S, \omega) + \int_0^S T(\mathbf{x}_Z) L_i(\mathbf{x}_Z, \omega) dz, \quad (1)$$

where $T(\mathbf{x}_S) = e^{-\sigma_t S}$ is the transmittance or ratio of radiance lost between the sensor and point $\mathbf{x}_S$, L_s is the radiance at the surface (point $\mathbf{x}_S$), and L_i represents the *in-scattering*: all incident irradiance at any point in the medium scattered towards the sensor. Finally, the integral sums the contribution of every point $\mathbf{x}_Z$ along the path.

The fundamental difficulty of this problem is the computational cost of approximating this integral, as the in-scattering itself is yet another nested integral over all directions Ω :

$$L_i(\mathbf{x}, \omega) = \int_{\Omega} \sigma_s f_s(\mathbf{x}, \omega_i, \omega) L(\mathbf{x}, \omega_i) d\omega_i, \quad (2)$$

where $f_s(\mathbf{x}, \omega_i, \omega)$ is the phase function, which represents the ratio of light being scattered from ω_i towards ω .

2. In-scattering approximation

In this work, as illustrated in Figure 1, we propose to separate in-scattering into two components: multiple scattering (which occurs when light is scattered several times in the medium before being in-scattered) and single scattering (light from the surface is directly in-scattered). This allows us to combine an industry standard approximation for single scattering, froxel lighting [Vos14, Hil15, Kov20], with a novel multiple scattering approximation derived from oceanography.

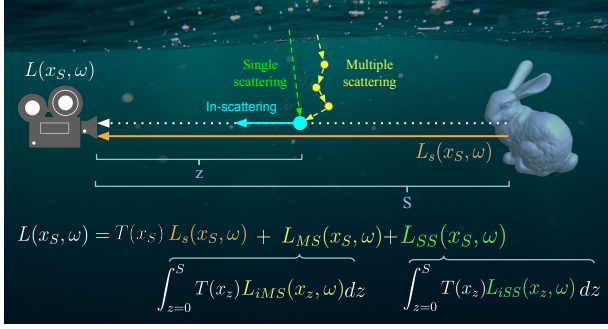


Figure 1: The final radiance L results from the contributions of the surface L_s , and the medium in-scattering. In-scattering is composed of multiple scattering L_{MS} and single scattering L_{SS} .

Multiple scattering As multiple scattering is prohibitively expensive to evaluate in real-time, we leverage real-world measurements to approximate it at any depth. In oceanography, the diffuse downwelling attenuation coefficient $K_d(\lambda)$ measures the attenuation of irradiance in the water volume per meter for wavelength λ [Mob94, LDC*05, Mob01, Gor89].

This coefficient relates the irradiance at the surface of the water body $E_{D_0}(\lambda)$ to the irradiance $E_D(\lambda, y)$ at any given depth y :

$$E_D(\lambda, y) = E_{D_0}(\lambda) e^{-K_d(\lambda)y}. \quad (3)$$

Substituting the in-scattering from Figure 1 for this, $L_{iMS}(y) \approx E_D(y)$, allows us to analytically solve the integral over the whole path L_{MS} :

$$\begin{aligned} L_{MS}(\mathbf{o} + S\omega, \omega) &= \int_0^S T(\mathbf{x}_S) \sigma_s f_s(\mathbf{x}, \omega_i, \omega) E_D(y) dz \\ &= \int_0^S e^{-\sigma_t z} \frac{\sigma_s}{4\pi} E_{D_0} e^{-K_d(o_y - z\omega_y)} dz \\ &= \frac{\sigma_s e^{-K_d o_y} E_{D_0}}{4\pi(K_d \omega_y - \sigma_t)} (e^{(K_d \omega_y - \sigma_t)S} - 1) \end{aligned} \quad (4)$$

where we use the isotropic phase function, $1/4\pi$. Note that this approximation has turned two nested integrals into a single $O(1)$ expression for each wavelength, which allows to evaluate it in real time per pixel.

Spectral rendering Equation (4) is computed per wavelength band and results in spectrum $L(\lambda)$. To convert it to RGB simulating different sensors, we compute the inner product between the spectrum and the sensor response curve $f(\lambda)$:

$$L_{RGB} = \int_{\lambda} L(\lambda) f(\lambda) d\lambda \approx w \sum_{\lambda} L(\lambda) f(\lambda), \quad (5)$$

where w is the width of the bands: $w = \frac{\max(\lambda) - \min(\lambda)}{n}$. We approximate this integral by applying the rectangle rule, which discretizes the problem.

Single scattering To add realism, L_{SS} (Figure 1) can be approximated separately and summed to the other radiance. In this work, we have integrated our L_{MS} approximation into the High Definition Rendering Pipeline (HDRP) [Lag18], in the Unity Engine. This allows us to utilize HDRP's implementation of froxel lighting.



Figure 2: Same underwater scene (JerlovB) rendered with a path tracer (left, 4 h) and our approximation (right, 2 ms).

3. Results

Figure 2 shows an example validation scene with the same physical parameters rendered with a volumetric spectral path tracer (left), and our real-time method (right). In this scene, as well as in others with a wide variety of coefficients and conditions, our method achieves similar underwater appearance at a fraction of the computational cost (20 ms instead of 4 h).

Together with the single scattering term, our pipeline can model volumetric effects such as spotlights or caustics, offering a richer underwater appearance, as we can see in Figure 3.



Figure 3: Examples showcasing the full pipeline joining both multiple and single scattering.

4. Conclusions

Our proposed method achieves a real-time frame rate rendering a wide variety of spectral underwater scenes. We have integrated our method into the Unity engine HDRP to combine it with some of its physically-based rendering features, and provide an interactive demo application.

There is a trade-off between speed and physical accuracy in the number of wavelength bands. The proposed approximation of the complex phenomenon of multiple scattering is limited by modeling downwelling irradiance as a function of the distance to the surface. We assume homogeneous coefficients and an isotropic phase function.

Tackling these limitations would be a promising research direction moving forward. As our model has been formally derived from radiative transfer theory and oceanography, it could represent a sound foundation for further development.

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